Digital Communication IC Design

Lecture 5: Scrambler, Descrambler and Self-Synchronization

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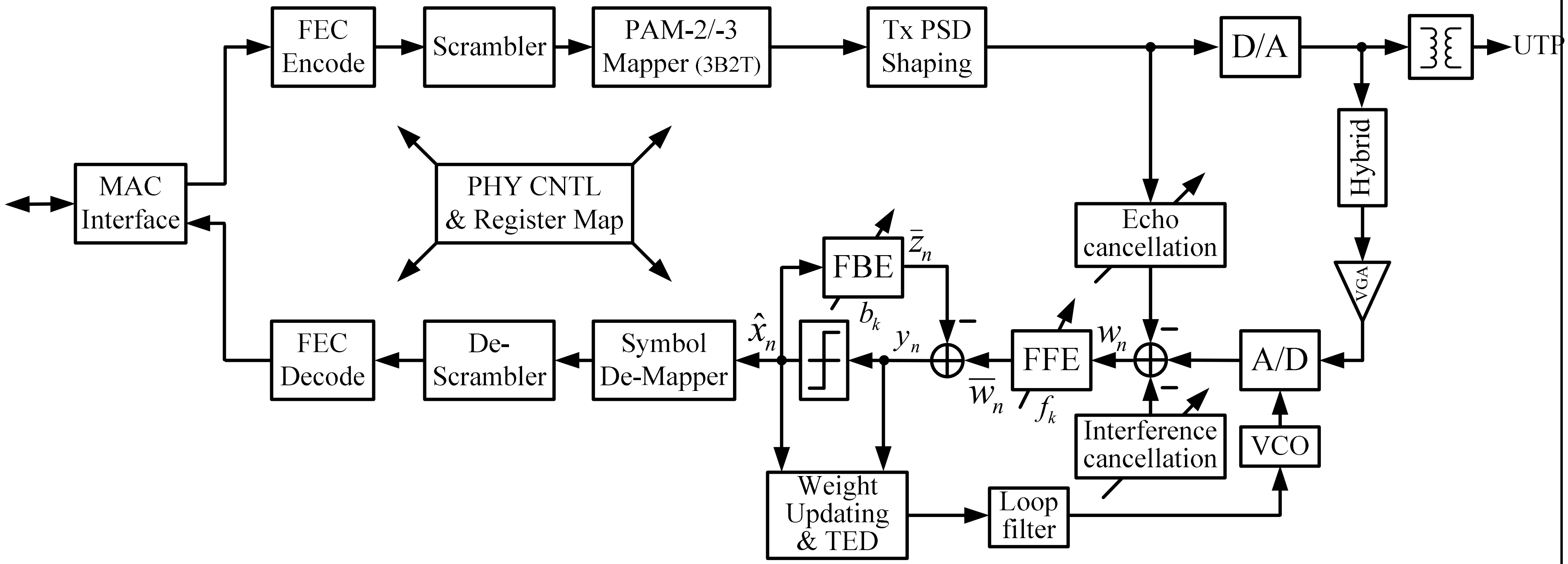
Apr. 15, 2026

Outline

- Pseudorandom Sequence
- Self-Synchronization Scrambler/Descrambler

Block Diagram of Automotive Ethernet 1000Base-T1 PHY



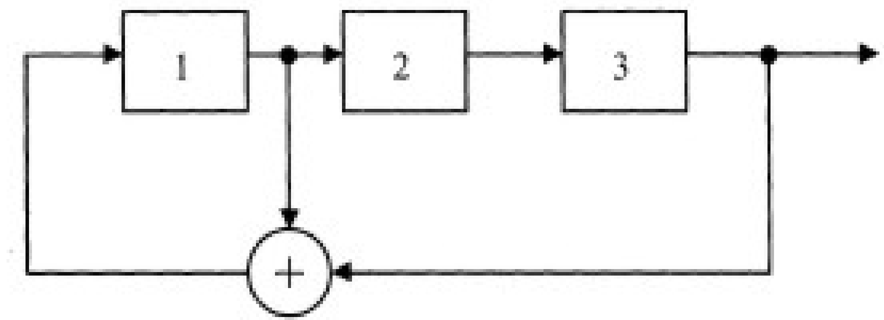
Scrambler

- Objective
  - Scrambling the data can minimize long strings of 0's and 1's and suppress discrete spectral components.
  - Scrambling devices randomize or "whiten" the data by producing digits that appear to be independent and equiprobable
- Two basic classes of scramblers
  - Scramble via logic addition (modulo 2) of a pseudorandom sequence
  - Generalize to M-ary transmission by using modulo M addition

[1]

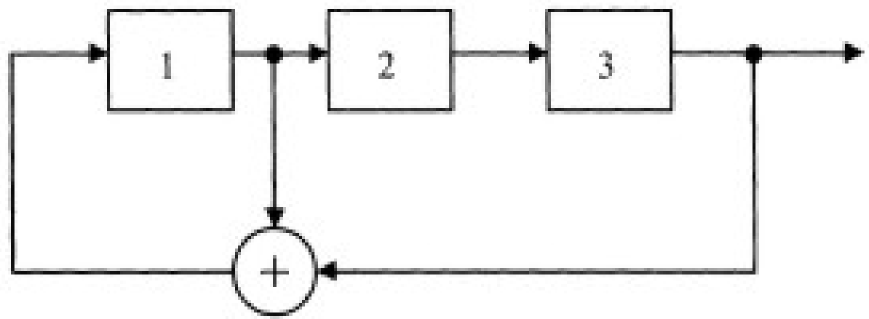
Pseudorandom Sequences

- Scramble via logic addition (modulo 2) of a pseudorandom sequence
  - Use shift registers with certain feedback connection to modulo 2 adders
    - Shift registers: a number of flip-flops (D-FF)
    - Feedback connection: consist of taps at certain stages
    - The tapped signal added modulo 2 and fed back to the first FF
- 3-stage sequence generator
  - $f(x) = x^3 + x + 1$



Pseudorandom Sequences

- The length of a pseudorandom sequences
  - Determined by the shift register length, the feedback taps and the initial states of the flip-flops
  - For example, $f(x) = x^3 + x + 1$
    - Maximum-length sequences: $2^n - 1$ (n: the no. of stages for shift register)

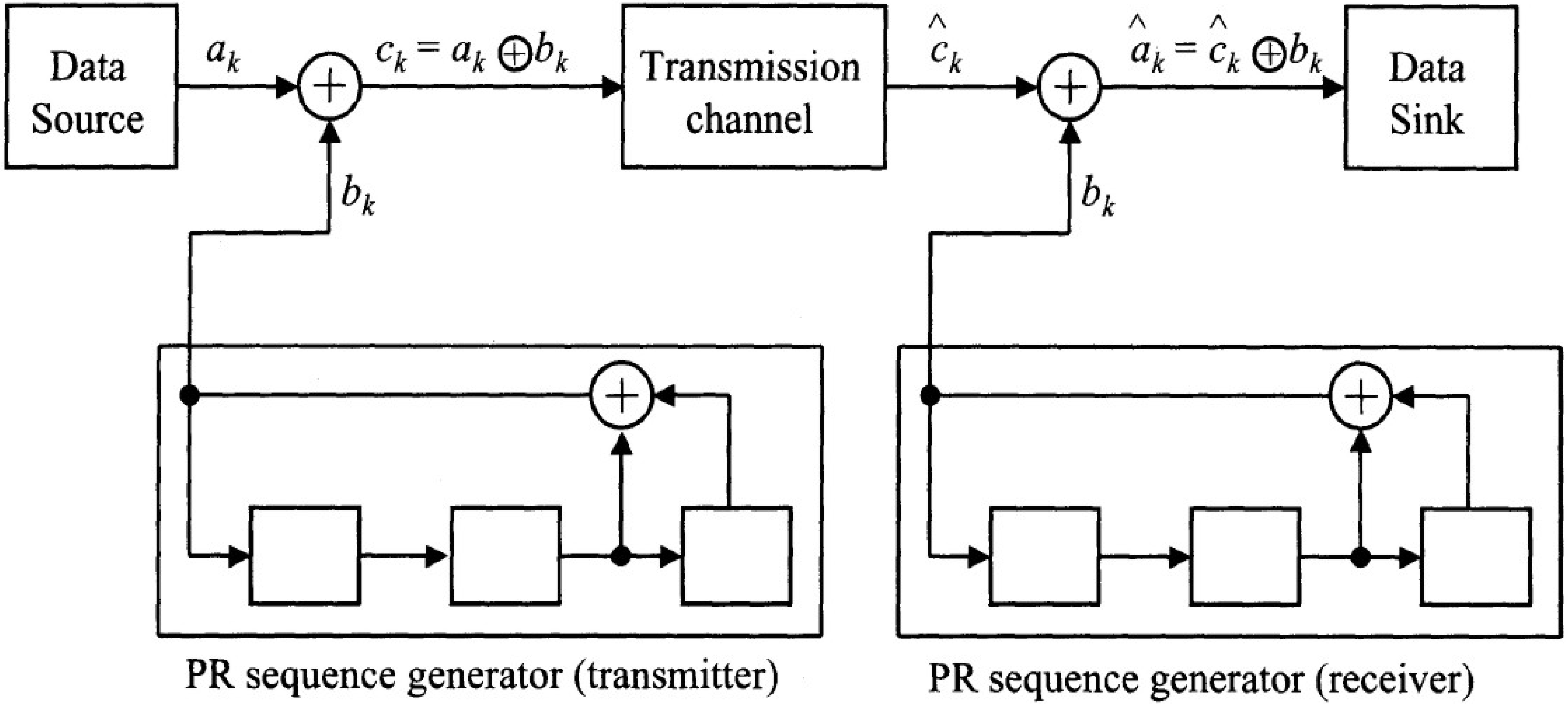


	Stage			
Time	1	2	3	
1	1	1	1	← Initial condition
2	0	1	1	
3	1	0	1	
4	0	1	0	
5	0	0	1	
6	1	0	0	
7	1	1	0	
8	1	1	1	

Maximum-Length of Various Pseduorandom Sequences

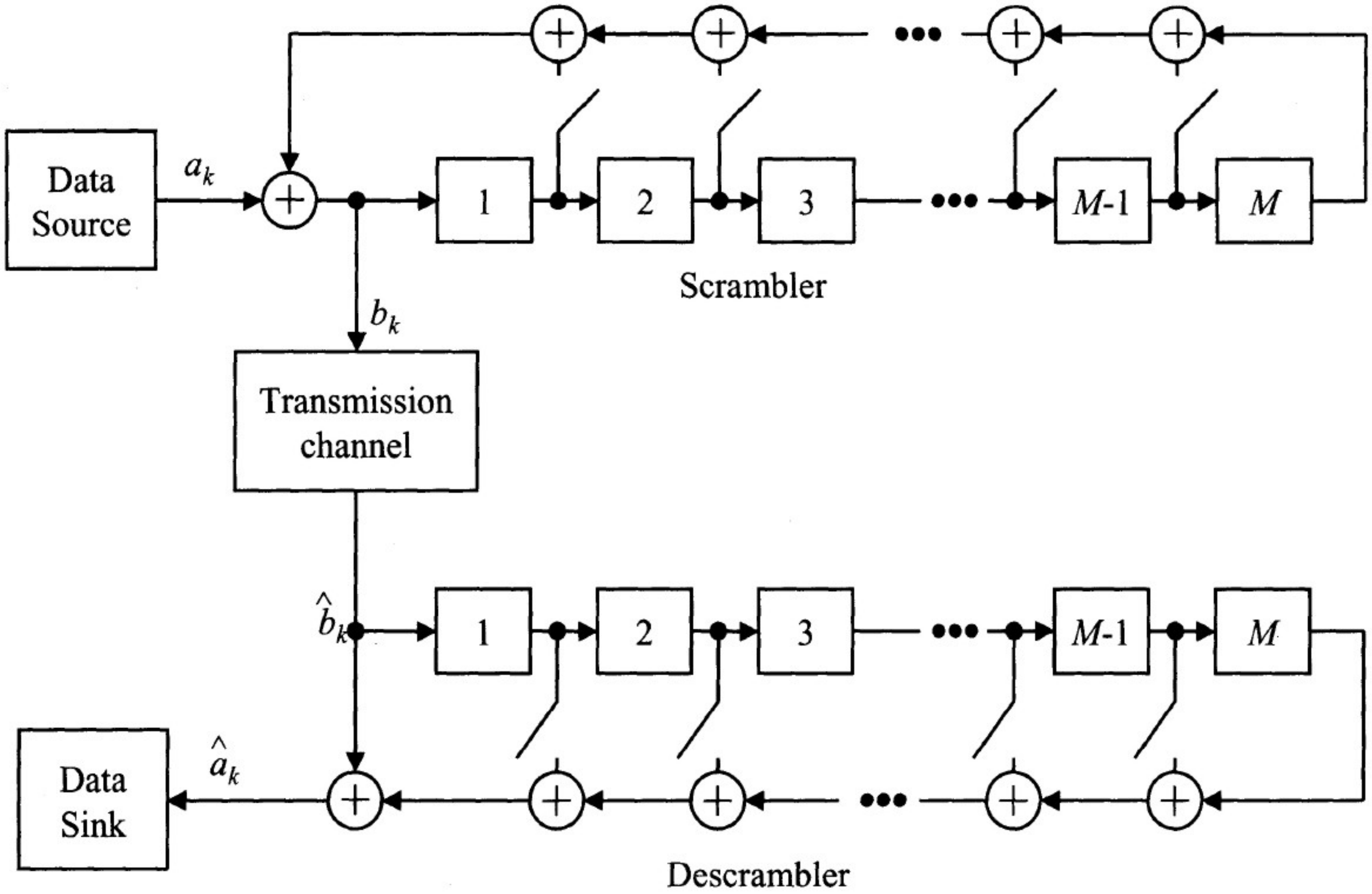
Length of Shift Register	Feedback Taps	Period of Sequence
3	1,3	7
4	1,4	15
5	2,5	31
6	1,6	63
7	1,7	127
8	1,6,7,8	255
9	4,9	511
10	3,10	1,023
11	2,11	2,047
12	2,10,11,12	4,095
13	1,11,12,13	8,191
14	2,12,13,14	16,383
15	14,15	32,767
16	11,13,14,16	65,535
17	14,17	131,071
18	11,18	262,143
19	14,17,18,19	524,287
20	17,20	1,048,575

Scrambler and Descramber in Transceiver



- ❑ Scrambler in transmitter and descrambler in receiver
- ❑ Self-synchronization scrambler/descrambler

Generalized Self-Synchronization Scrambler



Generalized Self-Synchronization Scrambler

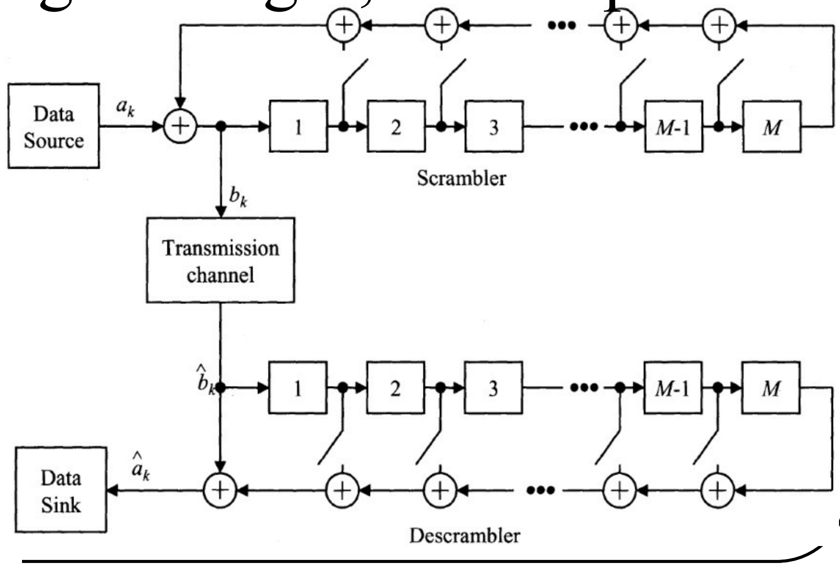
- For a scrambler containing M stages, the output of the transmitted bit b_k

$$b_k = a_k \oplus \sum_{p=1}^{M} b_{k-p}\delta_p,\ \delta_p = \begin{cases} 1 & \text{if stage } p \text{ is fedback and added} \\ 0 & \text{otherwise} \end{cases}$$

  - a_k: the transmitted source bit-stream
  - $\oplus$ and $\sum$: the modulo addition

- For a descrambler (decoder) having M stages, the output of the recovered source bit-stream

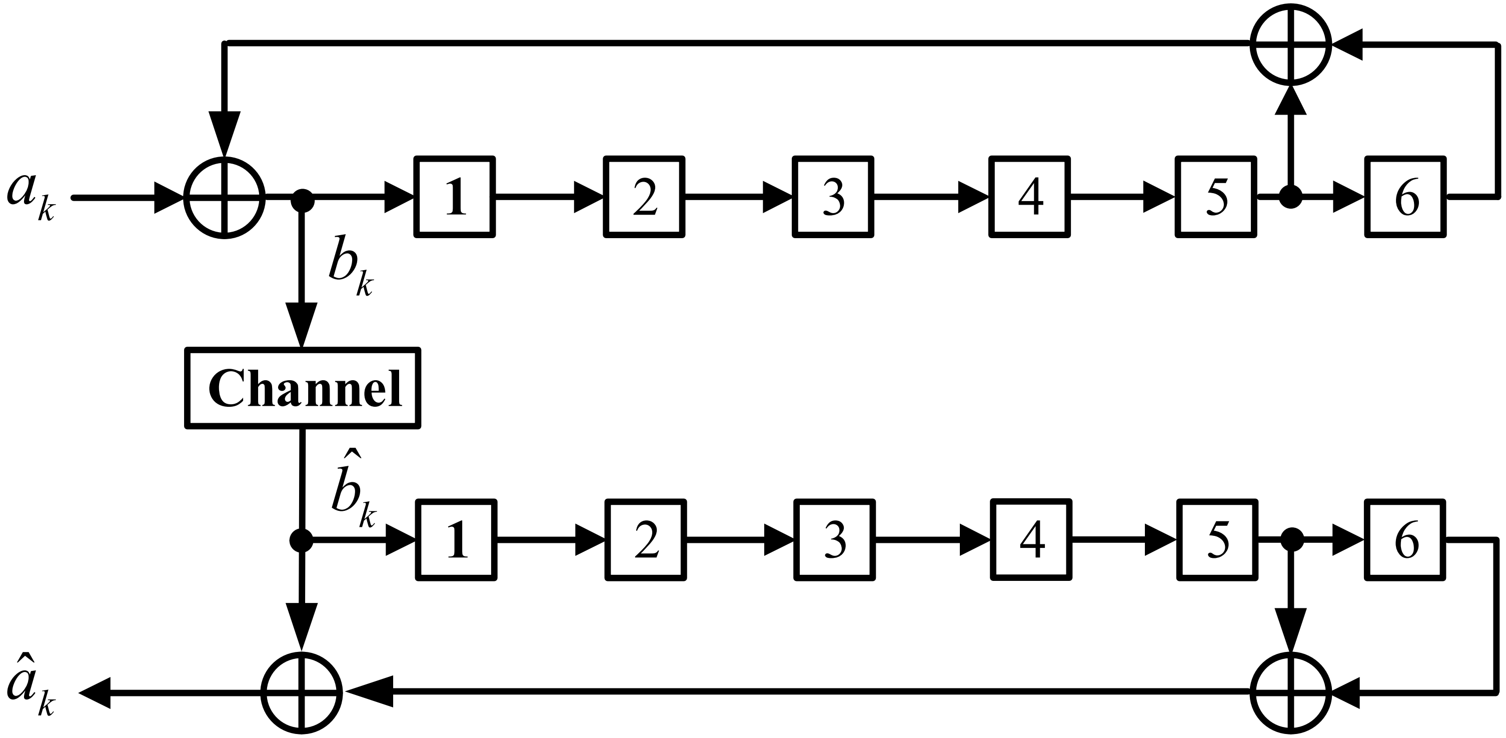
$$\hat{a}_k = \sum_{p=0}^{M} \hat{b}_{k-p}\delta_p,$$



Scrambler/Descrambler $f(x) = x^6 + x^5 + 1$

- For a six-stage self-synchronization scrambler, let δ_5 and $\delta_6 = 1$ and all other feedback connections $\delta_i = 0$ for $i = 1\ to\ 4$. The scrambled data sequence is

$$b_k = a_k \oplus b_{k-5} \oplus b_{k-6}$$



Proof

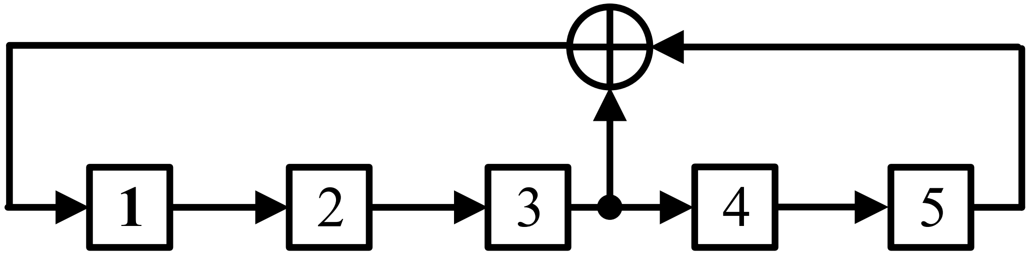
- Assuming error-free transmission (i.e., $\hat{b}_k = b_k$), the descrambled data sequence is

$$\hat{a}_k = \hat{b}_k \oplus b_{k-5} \oplus b_{k-6} = (a_k \oplus b_{k-5} \oplus b_{k-6}) \oplus b_{k-5} \oplus b_{k-6} = a_k$$

- The descrambled sequence is identical to the original data sequence.

Demonstration

- ❑ Scrambler: $f(x) = x^5 + x^3 + 1$

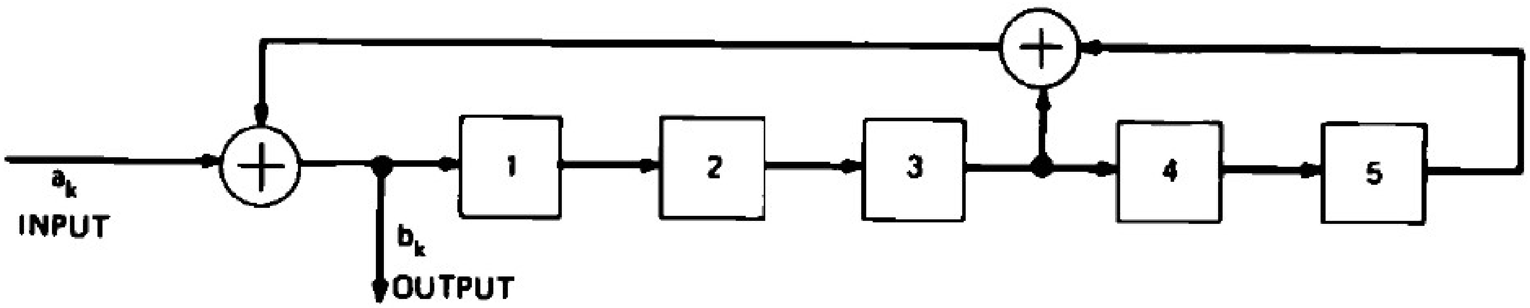


- ❑ Please see the demo in class

```
% Initial Value
% (1) (2) (3) (4) (5) -- Tap !!!
% 1   0   0   0   0    -- intial value !!!
  0   1   0   0   0   % 1
  0   0   1   0   0   % 2
  1   0   0   1   0   % 3
  0   1   0   0   1   % 4
  1   0   1   0   0   % 5
  1   1   0   1   0   % 6
  0   1   1   0   1   % 7
  0   0   1   1   0   % 8
  1   0   0   1   1   % 9
  1   1   0   0   1   % 10
  1   1   1   0   0   % 11
  1   1   1   1   0   % 12
  1   1   1   1   1   % 13
  0   1   1   1   1   % 14
  0   0   1   1   1   % 15
  0   0   0   1   1   % 16
  1   0   0   0   1   % 17
  1   1   0   0   0   % 18
  0   1   1   0   0   % 19
  1   0   1   1   0   % 20
  1   1   0   1   1   % 21
  1   1   1   0   1   % 22
  0   1   1   1   0   % 23
  1   0   1   1   1   % 24
  0   1   0   1   1   % 25
  1   0   1   0   1   % 26
  0   1   0   1   0   % 27
  0   0   1   0   1   % 28
  0   0   0   1   0   % 29
  0   0   0   0   1   % 30
  1   0   0   0   0   % 31
  0   1   0   0   0   % 32
```

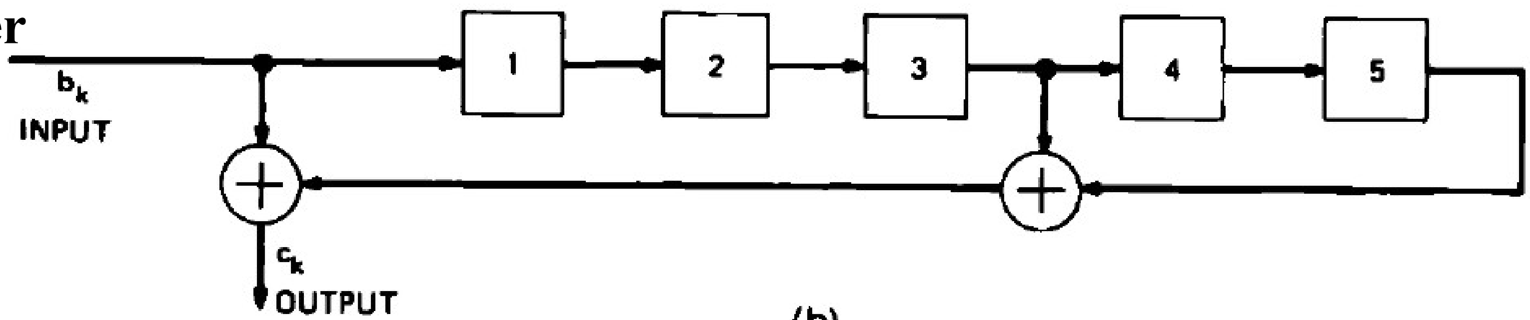
Self-Synchronization Scrambler/Descrambler

Scrambler

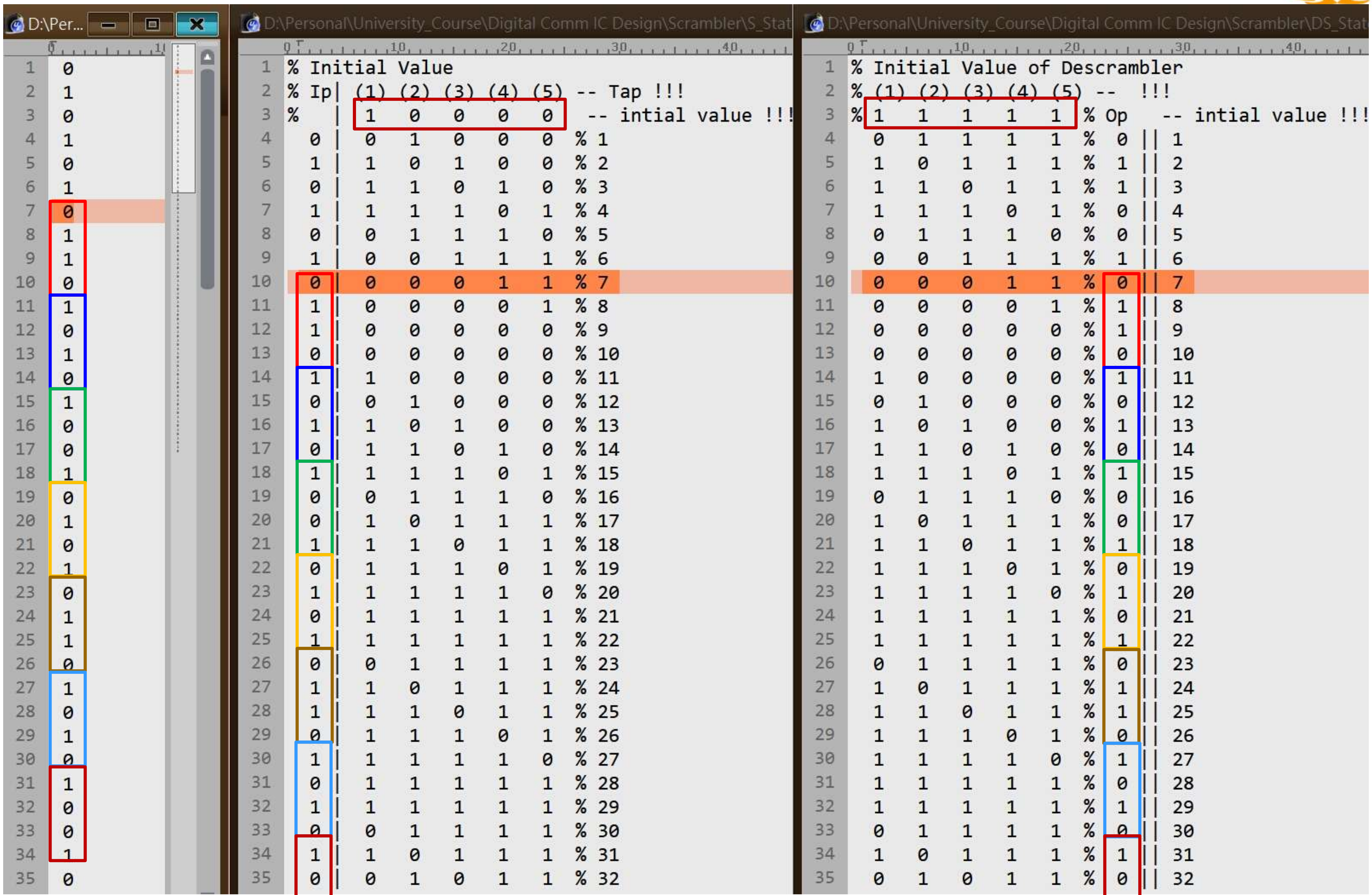


(a)

Descrambler



(b)



D:\Per...

```
0
1
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0
0
1
0
```

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```
% Initial Value
% Ip| (1) (2) (3) (4) (5) -- Tap !!!
%   |  1   0   0   0   0   -- intial value !!!
  0 |  0   1   0   0   0  % 1
  1 |  1   0   1   0   0  % 2
  0 |  1   1   0   1   0  % 3
  1 |  1   1   1   0   1  % 4
  0 |  0   1   1   1   0  % 5
  1 |  0   0   1   1   1  % 6
  0 |  0   0   0   1   1  % 7
  1 |  0   0   0   0   1  % 8
  1 |  0   0   0   0   0  % 9
  0 |  0   0   0   0   0  % 10
  1 |  1   0   0   0   0  % 11
  0 |  0   1   0   0   0  % 12
  1 |  1   0   1   0   0  % 13
  0 |  1   1   0   1   0  % 14
  1 |  1   1   1   0   1  % 15
  0 |  0   1   1   1   0  % 16
  0 |  1   0   1   1   1  % 17
  1 |  1   1   0   1   1  % 18
  0 |  1   1   1   0   1  % 19
  1 |  1   1   1   1   0  % 20
  0 |  1   1   1   1   1  % 21
  1 |  1   1   1   1   1  % 22
  0 |  0   1   1   1   1  % 23
  1 |  1   0   1   1   1  % 24
  1 |  1   1   0   1   1  % 25
  0 |  1   1   1   0   1  % 26
  1 |  1   1   1   1   0  % 27
  0 |  1   1   1   1   1  % 28
  1 |  1   1   1   1   1  % 29
  0 |  0   1   1   1   1  % 30
  1 |  1   0   1   1   1  % 31
  0 |  0   1   0   1   1  % 32
```

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```
% Initial Value of Descrambler
% (1) (2) (3) (4) (5) --  !!!
% 1   1   1   1   1   % Op  -- intial value !!!
  0   1   1   1   1   %  0 || 1
  1   0   1   1   1   %  1 || 2
  1   1   0   1   1   %  1 || 3
  1   1   1   0   1   %  0 || 4
  0   1   1   1   0   %  0 || 5
  0   0   1   1   1   %  1 || 6
  0   0   0   1   1   %  0 || 7
  0   0   0   0   1   %  1 || 8
  0   0   0   0   0   %  1 || 9
  0   0   0   0   0   %  0 || 10
  1   0   0   0   0   %  1 || 11
  0   1   0   0   0   %  0 || 12
  1   0   1   0   0   %  1 || 13
  1   1   0   1   0   %  0 || 14
  1   1   1   0   1   %  1 || 15
  0   1   1   1   0   %  0 || 16
  1   0   1   1   1   %  0 || 17
  1   1   0   1   1   %  1 || 18
  1   1   1   0   1   %  0 || 19
  1   1   1   1   0   %  1 || 20
  1   1   1   1   1   %  0 || 21
  1   1   1   1   1   %  1 || 22
  0   1   1   1   1   %  0 || 23
  1   0   1   1   1   %  1 || 24
  1   1   0   1   1   %  1 || 25
  1   1   1   0   1   %  0 || 26
  1   1   1   1   0   %  1 || 27
  1   1   1   1   1   %  0 || 28
  1   1   1   1   1   %  1 || 29
  0   1   1   1   1   %  0 || 30
  1   0   1   1   1   %  1 || 31
  0   1   0   1   1   %  0 || 32
```

References

[1] David R. Smith, "Digital Transmission Systems," vol. II, 3rd ed., 2004, Springer S+Business Media New York.

[2] D. G. Leeper, "A Universal Digital Data Scrambler," The Bell Sys. Tech. Journal, vol. 52, no. 10, Dec. 1973, pp. 1851-1865.